# Do the right thing.
## Enhancing Research Practices through Data Citation

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# Introduction

CESSDA Data Citation Working Group

The Importance of Data Citation

Main Goals of the Guide

Core Elements of Data Citation

Recommendations for Key Stakeholders

Initial Survey Results

# Data Citation Guide

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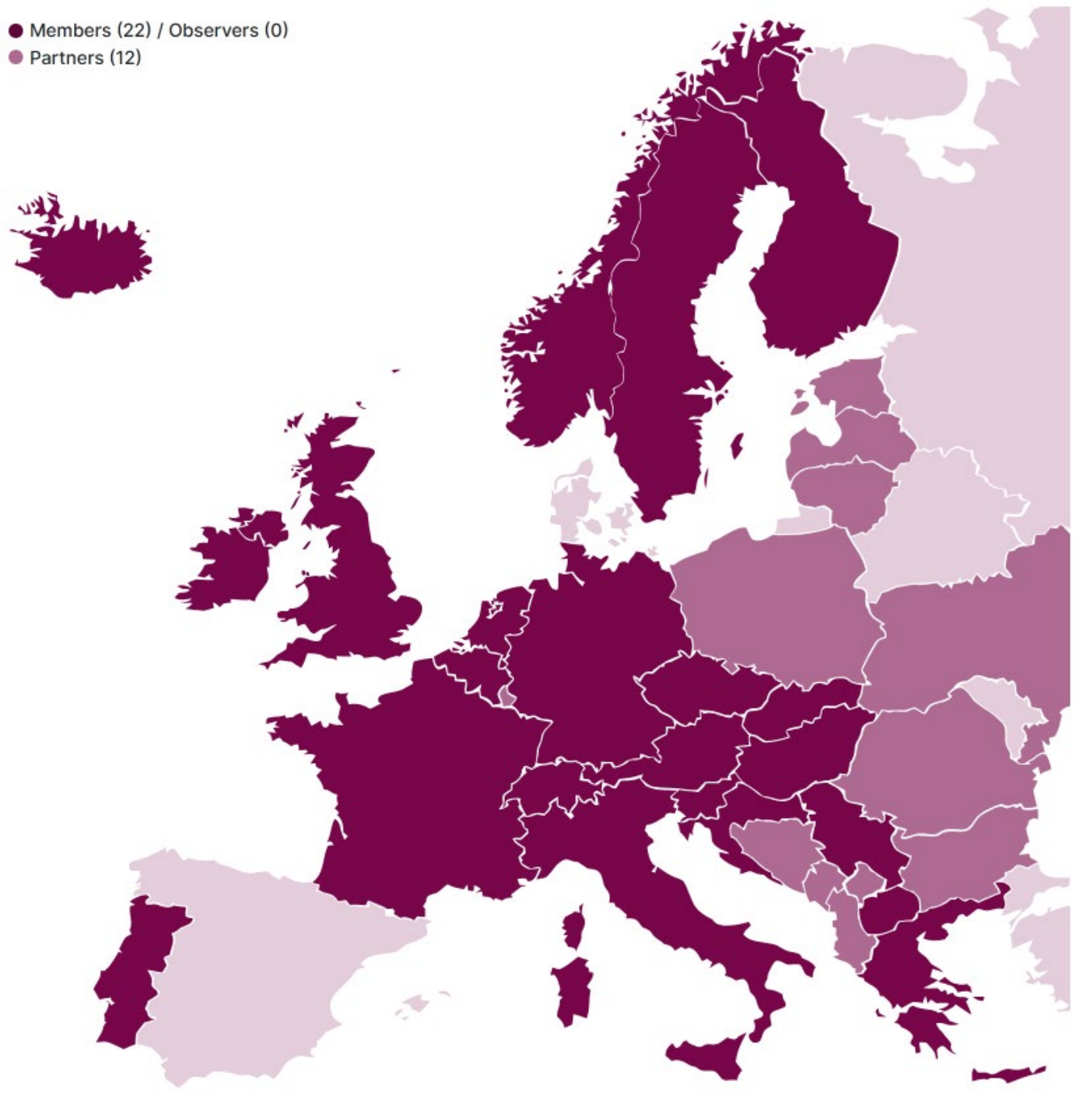


CESSDA, Consortium of European Social Science Data Archives represents the European Research Infrastructure Consortium.

Offers extensive, integrated, and sustainable data services to the social sciences.

Promotes social science research results and support national and international research and cooperation.

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# DATA CITATION WORKING GROUP

Started as CESSDA's collaborative mentorship* initiative, 2023

- Formulate (public) data citation recommendations targeting repositories, authors and journals and publishers
- Plan a roadmap of activities around data citation

Broadened the scope to cover various other stakeholders, 2024

- Finalise the recommendation and survey CESSDA Service Providers on their practices supporting data citation

Publish the recommendation and the dedicated website, 2025

- Ongoing outreach efforts

Collaboration

- FORS (lead), SND, FSD, Sikt, CSDA, ADP, So.Da.Net, TARKI, UKDS, CROSSDA… and feedback from the whole CESSDA Service Provider community (In-kind contribution)

* Encourages *collaboration* between SPs beyond official CESSDA projects and activities, to explore new avenues of cooperation which could fill gaps within the institutions and/or foster innovation and progress.

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# WHY CITE DATA?

A standard scholarly practice for independent research outputs

- Yet fairly uncommon in practice for both primary and secondary data use

+ Accountability
  - Acknowledges the effort involved in data production and curation (data ecosystem)
  - Ensures the long-term addressability of data (persistent identifiers)
+ Reproducibility
  - Ensures discoverability -> a precondition for reproducibility
  - Facilitates transparency through enhancing visibility and linkage
+ Policy Compliance
  - Required to fulfill funder, publisher or RPO policies
+ Reuse (of metadata and data)
  - Promotes reuse of data through enhanced visibility
  - Associated metadata helps to explore new research questions
  - Facilitates machine-actionability
  - Pivotal for knowledge graphs and connecting data, publications, authors, and funding

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# MAIN GOAL

Not to tell that data need to be cited because… data citation important

- It is, but..

Recommendations aim to **enhance** and **facilitate** data citation **practices**

- By **enabling** and simplifying data citation
  - Identify core requirements
  - Advice on how proper data citation is possible though concrete actions and policies
  - Describe the scope of the data citation and tools/services needed (ecosystem)
- By enhancing data citation **interoperability** (FAIR)
  - Metadata, metadata, metadata, persistent identifiers, machine-actionability, services
  - Identify and cite data and replication materials
- By raising **awareness** among key stakeholder groups
  - Most recommended actions directed towards other stakeholders than researchers
  - Encourage dialogue and best practices

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# GENERAL RECOMMENDATION: CORE COMPONENTS OF A DATA CITATION

A data citation

- Is a reference to the data itself
- Appears in-text or as a footnote whenever data are mentioned
- Appears in the reference list

## The anatomy of a data citation

**Core components**

- Data author(s)
- Title of the data
- Year of publication
- Version
- Data publisher
- Persistent Identifier

**Additional components**

- Data number or accession number
- Resource type [Dataset, flexible]
- Place of publication

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# DATA CITATION IN ACTION

## Full reference for a data citation following APA7 bibliographic style:

Data author(s) (Publication year). Data title (Data number; Version) [Resource type]. Data publisher. Persistent identifier

## Examples

ISSP Research Group (2023*). International Social Survey Programme: Environment IV - ISSP 2020* (ZA7650; Version 2.0.0) [Dataset]. GESIS. https://doi.org/10.4232/1.14153

Haerpfer, C., Inglehart, R., Moreno, A., Welzel, C., Kizilova, K., Diez-Medrano, J., Lagos, M., Norris, P., Ponarin, E. & Puranen B. (2022). *World Values Survey: Round Seven – Country-Pooled Datafile* (Version: 6.0) [Dataset]. World Values Survey Association. https://doi.org/10.14281/18241.18

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# STAKEHOLDERS IN DATA CITATION ECOSYSTEM

Targeted recommendations

- Data Repositories
- Authors
- Journals and Publishers
- Research Performing Organisations
- Research Policy and Strategy Development Entities
- Ethics Committees

1

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**Outreach**

Libraries and library networks, research and professional associations, citizen scientists, research funding organisations, reference style developers…

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# RECOMMENDATIONS
## FOR DIFFERENT STAKEHOLDER GROUPS

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# AUTHORS

1. **Cite the data themselves**
2. Include all six **core components** in the reference list
3. Cite **primary and secondary** data as well as replication materials

Explicitly cite all data used

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# JOURNALS AND PUBLISHERS

1. Allow data citations in the **standard list of references**
2. Require authors to cite data specifically
3. Include all **core components** in data citations and provide citation examples
4. Encourage authors to **publish data** and replication materials as independent publications
5. Encourage authors to provide **PIDs** for published data
6. Allow **time for quality assurance** and data curation

Allow proper citing of data as an independent output

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# RESEARCH PERFORMING ORGANISATIONS

1. Encourage data citation practices
2. Encourage authors to **report data publications** as research outputs
3. Include data publication **metadata in indexing** services
4. Include and leverage the **added value of metadata** describing data publications in administrative systems
5. Take data publications into account when summarising outputs

Index and take advantage of data publication metadata

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# HIGH-LEVEL RESEARCH POLICY AND STRATEGY DEVELOPMENT ENTITIES

1. Require data to be cited
2. Include all core components in guiding policies
3. Make the **benefits** of data citation **clear** in policies
4. Require **active reporting** of data publication outputs through research infrastructures

Require citing and reporting of data publications

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# ETHICS COMMITTEES

1. All research should follow **good citation practice**
2. Even **restricted** data should be **citable**
3. Address data citation in data management plans

Citing the work of others as a research ethical value

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# DATA REPOSITORIES

1. **Request data to be cited**
2. Provide all core data citation components
3. Provide data **versioning and access information**
4. Provide **citation suggestions** including the core components
5. Facilitate data citation through export and download of metadata for reference management software
6. Make citation information and other publication metadata accessible though generic **structured metadata** files
7. Make data publication and **relations** to other entities machine-actionable through rich metadata and **PIDs**
8. Raise **awareness** about data citation

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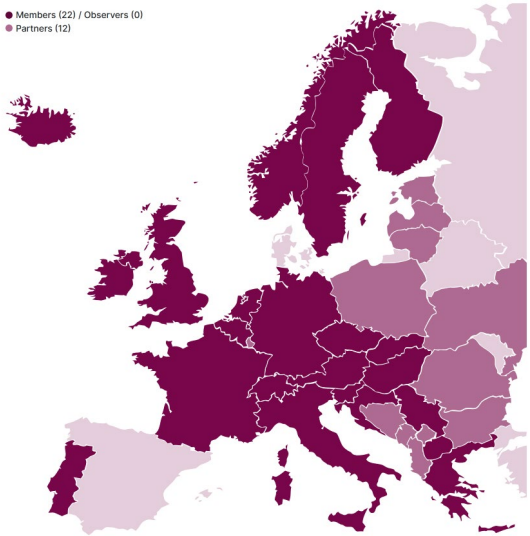
# CESSDA SERVICE PROVIDER SURVEY

**Universe**: All CESSDA Service Providers (members and partners = 34)

**Aim**: Assess their data citation practices and alignment with the recommendation (for repositories)

19 Service Providers answered the survey in 2024

- Final report is still being refined, and detailed results are CESSDA internal



# KEY FINDINGS

All SPs require data citation

- specificity of the requirement varies

Most core components are included in the provided (DDI) metadata

- Some do not provide publisher, year, and version

Most assign a Persistent Identifier (PID) prior to publication

Citation examples are mostly provided in the APA format

- Some include additional components, such as data producer/RPOs

Previous versions of datasets not always addressable

- A challenge as the goal is to be able to cite specific versions

Room for improvement in importing citation components into reference management software

More comprehensive information on the importance of data citation is needed to promote its adoption further!

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# CESSDA DATA CITATION GUIDE

datacitation.cessda.eu

Bornatici, C., Jernung, A., Alaterä, T. J., Tveit Sandberg, L., Strand, K., Štebe, J., & Trtíková, I. (2025). CESSDA Recommendations on Data Citation: Practical Recommendations for Key Stakeholders. https://doi.org/10.5281/zenodo.15043854

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# Thank you for your attention!

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